

# VoltGrid — SIH 2026 Slide 2: Proposed Solution Master Guide

**Official SIH Presentation Template — Slide 2:** Detailed explanation of the proposed solution, how it addresses the problem, and the innovation/uniqueness of the solution.

**Platform:** VoltGrid EV Mobility & Energy-Aware Routing Intelligence Platform.

**Reference Standard:** SIH 2026 Evaluation Rubric & PowerHouse Slide-by-Slide Defense Series.

## Executive Summary

Slide 1 established the team and project identity. **Slide 2 is where the jury decides whether to keep listening.** Its single objective is to establish:

*"We understand the real-world pain better than anyone else, we have a razor-sharp solution that can be pitched in 30 seconds, we have clear defensible USPs over existing tools, and we focus strictly on core features rather than feature-dumping."*

According to official SIH evaluators, teams fail Slide 2 by copy-pasting the portal problem statement, making generic claims ("efficient, smart, innovative"), and dumping 15 features.

Winning presentations use the **4 Master Points Flow**:

**PROVE (Research) → PITCH (30-Sec Idea) → ELEVATE (Defensible USP) → PROVE IT (Core Features Only)**

## 1. The 4 Master Points for Slide 2

| SLIDE 2: PROPOSED SOLUTION SKELETON                                                                                                                                                                                                                                           |                                                                                                                                                                                        |                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. PROVE THE PROBLEM <ul style="list-style-type: none"><li>Research-backed metrics</li><li>85% EVs are 2W in India</li><li>30-40% highway range loss</li><li>"Blind ETA" queue crisis</li></ul>                                                                               | 2. PITCH THE SOLUTION <ul style="list-style-type: none"><li>2-3 lines, pitch-style</li><li>Understandable in 30s</li><li>Active action verbs</li><li>Highlighted centerpiece</li></ul> | 3. ELEVATE WITH USP <ul style="list-style-type: none"><li>Clear "Why Us?" elevation</li><li>ETA Predictor vs Static Pin</li><li>Physics vs Generic GPS</li><li>Autonomous in-transit reroute</li></ul> |
| 4. PROVE IT WITH CORE FEATURES (LESS IS MORE) <ul style="list-style-type: none"><li>1. Energy-Aware Corridor Routing</li><li>2. Guaranteed Slot Booking &amp; QR Passes</li><li>3. Autonomous In-Transit Failover</li><li>4. Dual Driver &amp; B2B Operator Console</li></ul> |                                                                                                                                                                                        |                                                                                                                                                                                                        |

## 2. Presentation-Ready Slide Content (Copy-Paste for PowerPoint)

### A. Prove the Problem: Research-Backed Real Pain Points

*(Research beyond the SIH portal — proving you understand the operational bottleneck)*

- The 2W Highway Exclusion Bottleneck:** **85% of electric vehicles on Indian roads are 2-wheelers** with small 2.5–3.7 kWh battery packs *(Source: FADA / MoRTH 2024)*. Yet, over **90% of highway charging infrastructure is built exclusively for 4Ws (CCS-2 DC Fast)**, leaving light EVs stranded without 15A socket guidance or swapping kiosks.
- The Highway Physics Penalty:** Real-world highway range drops **30%–40% below laboratory ARAI claims** due to aerodynamic drag from pillion passengers, heavy cargo mass, and 38°C ambient battery heat. Standard navigation engines fail to compute these non-linear physical losses.

- The "Blind ETA" Availability Crisis: 28%–35% of highway charging points experience offline trips or 45+ minute queues (Source: SMEV Field Audits). Existing map apps show static charger status at departure ( $T=0$ ), which is completely obsolete by the time the driver arrives 90 minutes later ( $T = \text{ETA}$ ).

## B. Solution Pitch: The 30-Second Centerpiece

(Highlight this prominently in the center of Slide 2)

"VoltGrid is an energy-aware EV navigation and charging reservation platform that eliminates highway range anxiety by dynamically predicting battery depletion through first-principles physics and guaranteeing charger vacancy at arrival time across 925+ corridor stations."

## C. Elevate with USP: "Why Should the Jury Select Yours?"

(Structured as an elevation: Existing Limitation → VoltGrid Change → Resulting Value)

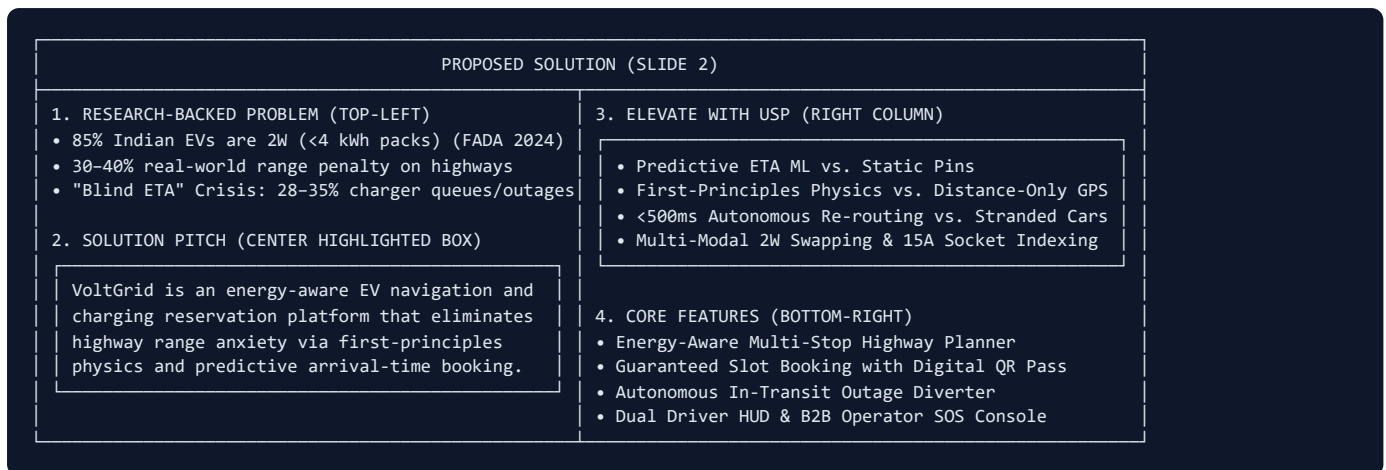
| DIMENSION       | EXISTING SOLUTIONS (GOOGLE MAPS, TATA EZ CHARGE, STATIQ)                                                  | VOLTGRID INNOVATION & ELEVATION                                                                                          | RESULTING VALUE                                                                                |
|-----------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1. Availability | <b>Static Point-in-Time Pins:</b> Shows if a charger is green right now at departure.                     | <b>Predictive ETA Availability Model:</b> Logistic regression model forecasting charger vacancy at future arrival time.  | <b>Zero Queue Surprise:</b> Eliminates arriving at an occupied 8-car queue or offline station. |
| 2. Energy Math  | <b>Distance-Only Routing:</b> Treats EV batteries like combustion petrol tanks.                           | <b>First-Principles Energy Core:</b> Models payload mass, pillion frontal drag (+15%), and road grade.                   | <b>Guaranteed Arrival SoC:</b> Driver reaches destination with configured safety buffer.       |
| 3. Reliability  | <b>Siloed &amp; Vulnerable:</b> Proprietary brand apps leave drivers stranded if a charger dies mid-trip. | <b>Autonomous In-Transit Recovery:</b> <code>/api/reroute</code> detects outages and recalculates backup hub in <500 ms. | <b>Fail-Safe Journey:</b> Dynamic recovery without user manual re-planning.                    |
| 4. Inclusivity  | <b>4W-Centric:</b> Completely ignores two-wheelers on intercity highway corridors.                        | <b>Multi-Modal Light EV Network:</b> Indexes 15A industrial plugs and battery swapping kiosks alongside DC plazas.       | <b>Inclusive Mobility:</b> Covers both commuter delivery 2Ws and luxury passenger 4Ws.         |

## D. Core Features: Only What Proves Feasibility (Less is More)

(Do NOT feature-dump; include only the 4 essential pillars)

1. **Physics-Backed Corridor Route Planner:** Pairwise highway routing (OSRM) with dynamic departure SoC targets and arrival reserve guarantees across Tamil Nadu & Bengaluru.
2. **Guaranteed Slot Reservation with QR Passes:** Advance bay booking with cryptographic Razorpay payment verification, eliminating 45-minute highway wait times.
3. **Autonomous Outage Recovery Engine:** In-motion monitoring that detects charger tripping and automatically diverts the vehicle to the next viable station in under 500 ms.
4. **Dual Driver HUD & B2B Operator Console:** Consumer navigation interface paired with an enterprise operator console for bay telemetry, load balancing, and emergency SOS dispatch.

### 3. Recommended Slide 2 Visual Layout (PowerPoint Wireframe)



### 4. Verbal Presentation Scripts for Slide 2

#### A. 30-Second Elevator Pitch for Slide 2

"Judges, over 85% of electric vehicles sold in India are two-wheelers with small 3 kWh batteries. Yet, existing map apps treat EVs like petrol cars, ignoring the fact that highway wind drag and passenger weight cut real-world range by over 30%. Even worse, map apps show if a charger is vacant right now—not when you arrive 90 minutes later.

**VoltGrid solves this:** We combine a first-principles thermodynamic energy model with a predictive ETA availability predictor across 925+ verified stations. If a charger goes offline while you are driving, our system autonomously recalculates an alternate route in under 500 milliseconds.

We turn EV intercity travel from an uncertain gamble into a predictable, zero-wait journey."

#### B. 60-Second Comprehensive Pitch for Slide 2

"Respected Evaluators, on Slide 2, we present our proposed solution, rooted in real-world field research.

Today, India has over 1.2 million electric two-wheelers. But our highway infrastructure is severely fragmented: 90% of chargers cater only to 4-wheelers, certified laboratory ranges drop by up to 40% on highways, and drivers face the 'Blind ETA Crisis'—arriving after a 90-minute drive only to find a 45-minute queue or a broken socket.

**Our solution is VoltGrid:** an energy-aware mobility platform that predicts physical battery consumption and reserves guaranteed charging bays at your arrival time.

## How do we elevate beyond existing solutions?

1. *Predictive vs. Static*: While Google Maps shows static pins, our machine learning model estimates occupancy at your future arrival time.

2. *Physics vs. Distance*: We calculate true consumption factoring in pillion drag, cargo mass, and temperature.

3. *Autonomous Recovery*: If your selected charger trips while you are on the highway, VoltGrid autonomously re-routes you in under 500 milliseconds.

4. *Light-EV Inclusivity*: We index 15A sockets and battery swapping kiosks specifically for two-wheelers.

VoltGrid is not just another map app; it is the operating system for reliable intercity EV mobility.\*\*

## 5. Pre-Submission Self-Audit Checklist

Before presenting Slide 2, verify your team satisfies all four criteria:

- ☒ **Problem proven by research** : Cited FADA/MoRTH 85% 2W market share and SMEV field reports (not repeating SIH portal text).
- ☒ **Pitch understandable in 30 seconds** : 2–3 line highlighted solution pitch using strong active verbs.
- ☒ **USP clearly elevated** : Distinct 4-point comparison contrasting existing limitations with VoltGrid's value.
- ☒ **Core features curated** : Restricted to 4 essential capabilities; zero feature-dumping; implementation details saved for Slide 3.